The improvement search: what improved the model, and what did not

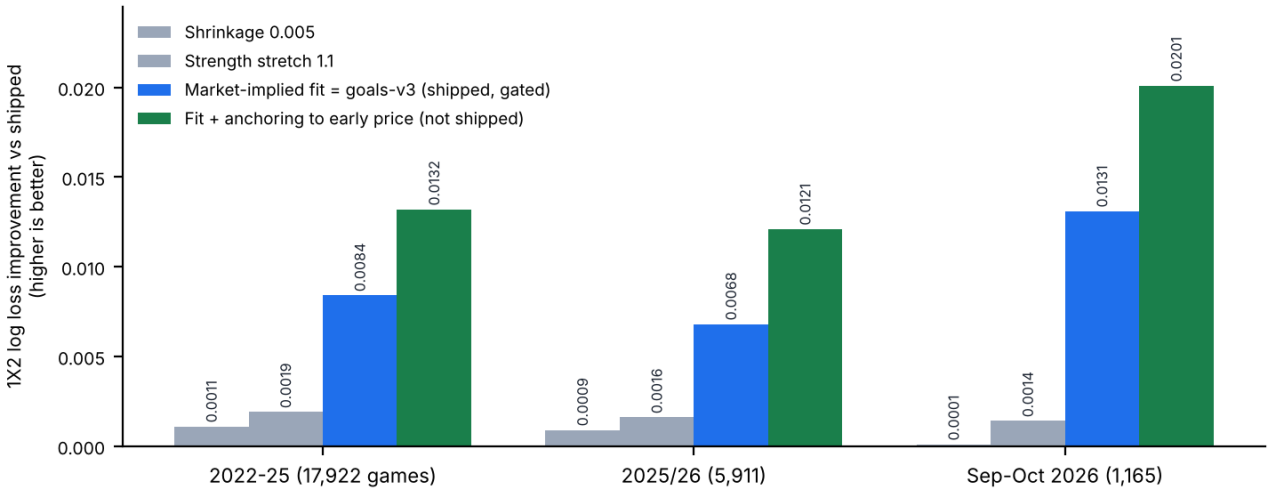
Prepared 6 November 2026. Merged to dev. Nothing published changes: the new model starts with games on 14 November and only after the data backfill.

Bottom line. One change clearly improves the model's probabilities: fitting each team's strength partly to the goals the bookmakers' closing prices implied for past games (**goals-v3**). The error in the home/draw/away chances falls by 0.008, 0.007 and 0.013 across three separate windows, each far outside the noise. **It does not raise the headline hit rate** (changes of +0.1, 0.0 and -0.2 points, inside the noise), because the pick thresholds were tuned to the old numbers; ROI at the closing price improves a little in all three windows. A second change, anchoring each game to the early-week price, adds a further gain and is held back until it has been checked on SportyBet's own prices.

1. What was tested

About sixty model variants, three ways of removing the bookmaker's margin, four ways of using the bookmaker's price, and the leading candidates run through the real pick layer. Every game is predicted only from results before its matchday, over 17 football-data divisions, on three windows: 2022-25 (17,922 games), 2025/26 (5,911) and Sep-Oct 2026 (1,165, your sample). Differences are paired on the same games with 95% intervals. The frozen holdout from 1 November was not touched.

2. What improved the probabilities



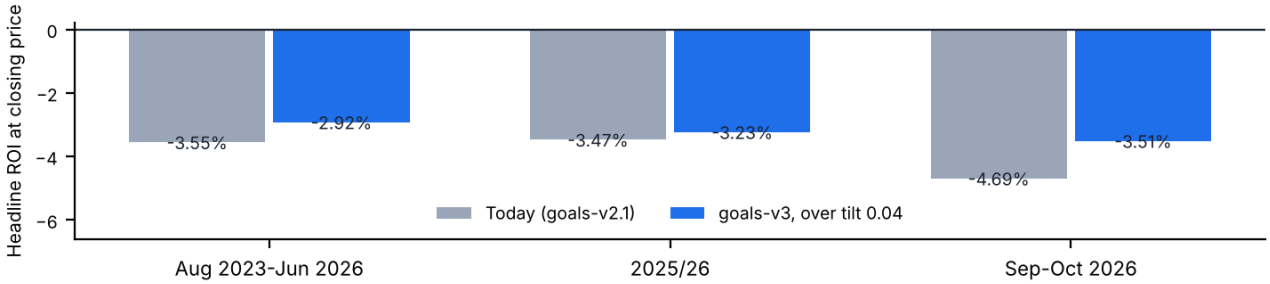
Bars are the fall in 1X2 log loss against the shipped model (the usual measure of how honest stated chances are). Every shipped bar's interval excludes zero. Over/under 2.5 improves too, by about 0.002 for goals-v3.

Why it works. A match is one noisy score drawn from a distribution. The closing price is the whole market's estimate of that distribution, with lineups and injuries inside it. Fitting 60% of each past game's scoring signal to the goals its closing 1X2 and over/under 2.5 imply (20% to shots on target, 20% to the score) leaves the ratings far less noise to carry. That lets the fit forget faster (a 120-day half-life, up from 180) with lighter shrinkage. The solver was checked against the lab on 760 games and matches to the last digit.

3. What it does to the picks

Run through the real pick layer (sweet-spot rules, headline floor and over tilt all live), 19,375 games from August 2023 to October 2026. "Today" is the model people see now.

	Headline hit vs today (95% interval)	ROI at closing price	Overs among headlines
Today	77.8%, 78.0%, 78.4%	-3.55%, -3.47%, -4.69%	29%, 30%, 37%
goals-v3, over tilt 0.02	+0.11 (-0.26 to +0.58), -0.17 (-1.02 to +0.53), -0.31 (-1.93 to +1.22)	-2.90%, -3.32%, -3.40%	25%, 25%, 33%
goals-v3, over tilt 0.04 (shipped)	+0.10, -0.01, -0.18	-2.92%, -3.23%, -3.51%	30%, 30%, 37%
goals-v3, over tilt 0.08 (your call)	+0.29 (-0.12 to +0.77), +0.26 (-0.54 to +1.00), -0.20	-2.68%, -2.86%, -2.77%	39%, 40%, 45%



Reading. The stated chances become more honest and the hit rate does not move. Hit rate is set by the pick layer and the price cap, and the sweet-spot thresholds were tuned to the old probabilities. With sharper probabilities the share of overs would have fallen from 29% to 25%, against your wish to favour overs, so goals-v3 ships with tilt 0.04, which restores today's share at the same hit rate. Tilt 0.08 gives about ten more points of overs with no measured cost, but every hit difference is inside its interval, so it is a product preference and not an accuracy gain. It is a one-line switch.

Where the pick layer re-derives its rules every month from the trailing two years of whatever probabilities feed it, hit rate does move, the same way in all three windows:

Rolling-rules hit rate	2023-25	2025/26	Sep-Oct 2026
Shipped model	76.9%	77.0%	77.6%
goals-v3 fit	77.3%	77.1%	78.1%
goals-v3 fit + anchoring 0.5	77.6%	78.0%	78.0%

That is +0.4 to +1.0 points against windows whose own error is about 0.4, 0.5 and 1.2 points: consistent in direction, not significant alone. It is the strongest argument for the monthly re-derivation below.

4. The bookmaker's price

SportyBet's early price is informative. On 538 finished games with price history, the de-vigged 1X2 scores 0.957 at 40 hours before kickoff and 0.954 at the last snapshot; over/under 2.5 scores 0.675 and 0.673. An early-week price is nearly as good as the closing one, which is what anchoring depends on. What is not yet shown is that pulling the model toward SportyBet's price improves the model on SportyBet's numbers; the anchoring gains above are on football-data prices.

Market on the SportyBet board	Margin	Note
1X2	4.5%	538 games with history: 3.9%
Over/under (2.5)	5.7%	538 games with history: 5.0%
Draw no bet	5.6%	Cheaper than Asian handicap 0 (6.0%) for the identical bet
Both teams to score	6.1%	
Double chance	6.3%	Level with Asian handicap +0.5, so no twin-market gain there
Asian handicap	6.0% to 6.5%	By line

The closing-price feed has stopped. The last closing price stored in the database is 19-20 Sep 2026 for 16 leagues and 28 Sep to 7 Oct for the other five, while results run to 4 Nov. It looks like a one-off backfill with nothing behind it (I did not check the ingestion host's scheduler). goals-v3 therefore reads the closing prices from the same football-data file row that carries each score, not from that table.

5. What did not work

• Half-life 90 or 720 days, shot weight 0 or 0.75, shrinkage 0.02, total or rho scaling, ensembles of half-lives: none better than the shipped fit on all three windows.

• xG, even with five seasons for the top five leagues: unstable between seasons, and it adds nothing once the fit uses the market's goals.

• Correcting the shape of the scoreline grid afterwards: worthless; the errors are in the means.

• Replaying Forge and Code Hub codes with the new probabilities: the tier moved 3 to 5 points either way run to run, so no conclusion can be drawn.

6. What was shipped, and how to switch it on

Everything is in **dev** (merge 45691e9). It does nothing until the steps below are done, and then it applies only to games kicking off from **00:00 Lagos on 14 November 2026** in leagues where at least 40% of the last year's training rows carry the market goals. National-team and cup pools never get it. The archive reader falls back to the old columns if the migration is late, so a missed step cannot turn into "no history".

Step	What	How
1	Deploy dev, then apply the migration	supabase/migrations/0102_market_implied_goals.sql (two columns, set_market_goals, extended upsert)
2	Backfill the 17 football-data leagues	/api/cron/refresh-archive?seasons=2526,2425,2324,2223
3	Backfill MLS, Liga MX, Argentina, Brazil	/api/cron/refresh-archive?only=market&marketDays=400&marketSkip=0, then 400 and 800
4	Check coverage	Share of historical_results rows from the last 365 days with home_mkt_goals set, by league (expect 90%+ for the 17)
5	Nothing else	Games from 14 Nov pick it up. If late, move MARKET_MODEL_FROM; a league under 40% coverage stays on today's model

Rollback: set MARKET_MODEL_FROM to a far date in src/lib/model/params.ts. No data needs undoing.

7. Not shipped, in order of value

• **Anchoring to the publish-time price.** A further gain of 0.005 to 0.007 on top of goals-v3. It needs the SportyBet board read before each prediction is built, in three places in service.ts. The model side is already written. Run it in shadow for three or four weeks against SportyBet's price history before enabling.

• **Monthly re-derivation of the sweet-spot rules.** +0.4 to +1.0 points of hit rate in the lab, and it would make any later model change safe to ship without hand-tuning thresholds.

• **Restart the closing-price feed** so the database is current again; it also serves the closing-line checks.

• **Draw no bet instead of Asian handicap 0** in Forge legs: half a point of margin.

Full evidence, the failed variants and the reproduction commands: section L of docs/model-audit-teams-2026-11.md.